

Online Appendix

Additional tables

Table A1: Summary statistics

	Mean	Min.	Max.	Obs.
<u>Covariates:</u>				
Female	0.23	0	1	1,417
Age (#)	40.17	16	99	1,417
Party member	0.82	0	1	1,417
Years of membership (#)	11.57	0	60	1,417
Experienced canvasser	0.37	0	1	1,417
Participated in workshop	0.22	0	1	1,417
Downloaded app	0.28	0	1	1,417
Prior belief about competitor	21.32	0	100	1,417
<u>Survey outcomes:</u>				
Posterior belief about competitor	17.91	0	100	1,406
Planned canvassing	0.53	0	1	1,402
Planned days (#)	4.13	0	60	1,402
<u>Behavioral outcomes:</u>				
Any canvassing	0.14	0	1	1,417
Canvassed days (#)	0.65	0	32	1,417
Canvassed doors (#)	32.36	0	1,045	1,417

Notes: Table 1 presents summary statistics for the full sample. Variables are given in percent, unless stated otherwise.

Table A2: Balance tests

	Treatment	Control	P-value (T = C)
Female	0.23	0.23	0.970
Age	40.44	39.92	0.604
Is party member	0.81	0.84	0.120
Years of party membership	11.72	11.43	0.698
Has experience canvassing	0.37	0.38	0.664
Participated in canvassing workshop	0.21	0.22	0.853
Downloaded app before survey	0.27	0.30	0.182
Has canvassed before survey	0.09	0.10	0.410
Days canvassed before survey	0.25	0.28	0.685
Doors visited before survey	10.20	11.06	0.892
Prior belief: competition	20.39	22.23	0.139

Notes: Table A2 presents all available pre-treatment covariates across the treatment and control group, the p -value is based on a t -test comparing the difference-in-means between the two groups. A regression of the treatment indicator on all covariates yields an insignificant p -value for the F-test of joint significance of 0.471.

Table A3: Treatment effects on app download

	Survey	App Data		
	click on app-link	within 24h	within one week	any time after survey
Treatment	0.007 (0.010)	-0.007 (0.007)	-0.004 (0.008)	-0.005 (0.012)
Observations	1,417	1,417	1,417	1,417
Control group mean	0.04	0.02	0.03	0.06

Notes: Table A3 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing all outcome variables related to app-download on the treatment dummy for the full sample. Treatment effects are obtained conditional on pre-specified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Table A4: Characteristics of app users

	App	No App	P-value(App = No App)
Female	0.22	0.24	0.510
Age	33.18	43.65	0.000
Is party member	0.89	0.81	0.003
Years of party membership	7.95	13.36	0.000
Has experience canvassing	0.46	0.33	0.000
Participated in door-to-door workshop	0.36	0.14	0.000
Belief: competition intentions	19.70	23.63	0.029
Belief: competition actual	13.46	16.40	0.044
Canvassing: yes	0.82	0.35	0.000
Canvassing: days	6.83	2.47	0.000

Notes: Table A4 presents differences between respondents who downloaded the app (256 supporters) and respondents who never downloaded the app (462 supporters) in the control group. p-values are based on robust standard errors.

Table A5: Treatment effects on behavior with additional controls - app user sample

	Any canvassing		Days canvassed		Door knocked	
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Without</i>	<i>With</i>	<i>Without</i>	<i>With</i>	<i>Without</i>	<i>With</i>
	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>
Treatment	-0.091** (0.046)	-0.080* (0.044)	-0.126 (0.421)	0.018 (0.417)	0.317 (21.833)	7.579 (21.545)
Control mean	0.462	0.462	2.052	2.052	98.466	98.466
Observations	465	465	465	465	465	465
Controls	No	Yes	No	Yes	No	Yes

Notes: Table A5 presents coefficients, robust standard errors (in parentheses) and randomization inference based p-values (in brackets) of a linear model (OLS), regressing the indicated behavioral outcomes on the treatment dummy for the sample with valid location data (97% of people that downloaded the application). Columns 1, 3, and 5 show the specification without control variables. Columns 2, 4, and 6 include all pre-specified control variables and a measure of marginality of the district (a dummy whether the vote share difference between the party and its strongest competitor is within five percentage points). Outcome data are obtained through a smartphone application used by the canvassers to register knocked doors. “Any canvassing” refers to any recorded canvassing activity. “Days canvassed” refers to the number of canvassed days. “Doors knocked” refers to the number of knocked doors winsorized at the 99th percentile. *p<0.1; **p<0.05; ***p<0.01.

Table A6: Treatment effects using logit estimation

	Any canvassing	
	(1)	(2)
Treatment	-0.389** (0.156)	-0.339* (0.190)
Marginal effect at mean	-0.046	-0.028
Control mean	0.160	0.160
Observations	1417	1417
Controls	No	Yes

Notes: Table A6 presents coefficients of logit estimation with robust standard errors in parentheses. In columns 1 and 2 we estimate a logit regression of a dummy indicating any canvassing on the treatment dummy. Column 1 shows the specification without control variables. Column 2 includes all pre-specified control variables. Outcome data are obtained through a smartphone application used by the canvassers to register knocked doors. “Any canvassing” refers to any recorded canvassing activity. *p<0.1; **p<0.05; ***p<0.01.

Table A7: Impact on post-treatment beliefs

	Full sample		Underestimators		Overestimators	
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Without</i>	<i>With</i>	<i>Without</i>	<i>With</i>	<i>Without</i>	<i>With</i>
	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>
Treatment	5.193*** (0.954)	5.091*** (0.946)	7.898*** (0.730)	7.796*** (0.725)	-15.134*** (3.452)	-15.716*** (3.598)
Control mean	15.354	15.354	10.389	10.389	56.429	56.429
Observations	1406	1406	1261	1261	145	145
Controls	No	Yes	No	Yes	No	Yes

Notes: Table A7 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing the posterior beliefs on the treatment dummy. Columns (1) and (2) display the results for all respondents. Columns (3) and (4) display results for respondents who underestimated the information provided on competing party members' canvassing intentions. Columns (5) and (6) display results for respondents who overestimated the information provided on competing party members' canvassing intentions. Pre-specified control variables include: party membership, number of years of party membership, age, gender, whether a participant has participated in a canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Table A8: Heterogeneity by pre-treatment belief

	Any canvassing		Days canvassed		Door knocked	
	<i>Without Controls</i>	<i>With Controls</i>	<i>Without Controls</i>	<i>With Controls</i>	<i>Without Controls</i>	<i>With Controls</i>
Treatment	-0.048** (0.020)	-0.035** (0.017)	-0.136 (0.161)	-0.071 (0.149)	-5.214 (8.043)	-1.968 (7.491)
Over-estimator	-0.063* (0.038)	-0.039 (0.030)	-0.419** (0.194)	-0.305* (0.169)	-14.742 (15.238)	-8.482 (14.000)
Treatment $\times$ Over-estimator	0.016 (0.051)	0.020 (0.045)	0.161 (0.284)	0.246 (0.278)	14.850 (22.694)	19.294 (21.210)
Effect on over-estimator	-0.031 (0.047)	-0.014 (0.042)	0.025 (0.235)	0.175 (0.234)	9.636 (21.221)	17.326 (19.940)
Control mean	0.160	0.160	0.712	0.712	34.148	34.148
Observations	1,417	1,417	1,417	1,417	1,417	1,417
Controls	No	Yes	No	Yes	No	Yes

Notes: Table A8 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing the indicated outcomes on the treatment interacted with a dummy for overestimation. Pre-specified control variables include: party membership, number of years of party membership, age, gender, whether a participant has participated in a canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. The sample is restricted to the control group. *p<0.1; **p<0.05; ***p<0.01.

Table A9: Selection into post-election survey

	Responded to post-election survey	
	<i>Without Controls</i>	<i>With Controls</i>
Treatment	-0.007 (0.018)	-0.007 (0.018)
Control mean	0.143	0.143
Observations	1,406	1,406
Controls	No	Yes

Notes: Table A9 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing a dummy capturing participation in the post-election survey on the treatment dummy. Pre-specified control variables include: party membership, number of years of party membership, age, gender, whether a participant has participated in a canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Table A10: Selection into post-election survey (underestimators)

	Responded to post-election survey	
	<i>Without Controls</i>	<i>With Controls</i>
Treatment	-0.010 (0.020)	-0.010 (0.020)
Control mean	0.148	0.148
Observations	1,261	1,261
Controls	No	Yes

Notes: Table A10 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing a dummy capturing participation in the post-election survey on the treatment dummy. Pre-specified control variables include: party membership, number of years of party membership, age, gender, whether a participant has participated in a canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. The sample is restricted to underestimators. *p<0.1; **p<0.05; ***p<0.01.

Table A11: Balance in post-election survey

	Treatment	Control	P-value (T = C)
Female	0.18	0.25	0.274
Age	44.74	42.43	0.426
Is party member	0.85	0.84	0.878
Years of party membership	14.89	13.06	0.414
Has experience canvassing	0.45	0.45	0.954
Participated in door-to-door workshop	0.31	0.25	0.408
Downloaded app before survey	0.36	0.32	0.576
Has canvassed before survey	0.09	0.12	0.452
Days canvassed before survey	0.21	0.21	0.964
Doors visited before survey	2.31	3.66	0.612
Prior belief: competition	18.55	20.14	0.587

Notes: Table A11 presents all available covariates across the treatment and control group in the post-election survey, which was implemented two months after the election. We report the full sample. The p -value is based on a t -test comparing the difference-in-means between the two groups. A regression of the treatment indicator on all covariates yields an insignificant p -value for the F-test of joint significance of 0.801.

Table A12: Balance in post-election survey (underestimators)

	Treatment	Control	P-value (T = C)
Female	0.20	0.22	0.674
Age	44.41	42.14	0.452
Is party member	0.85	0.85	0.967
Years of party membership	15.22	12.90	0.316
Has experience canvassing	0.48	0.45	0.689
Participated in door-to-door workshop	0.31	0.26	0.387
Downloaded app before survey	0.35	0.32	0.675
Has canvassed before survey	0.08	0.12	0.426
Days canvassed before survey	0.22	0.21	0.960
Doors visited before survey	2.51	3.93	0.625
Prior belief: competition	13.88	16.11	0.293

Notes: Table A12 presents all available covariates across the treatment and control group in the post-election survey, which was implemented two months after the election. We report the underestimator sample. The p -value is based on a t -test comparing the difference-in-means between the two groups. A regression of the treatment indicator on all covariates yields an insignificant p -value for the F-test of joint significance of 0.767.

Table A13: Correlation between canvassing behavior and beliefs about competitor effort (control group)

	Any canvassing		Days canvassed		Door knocked	
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Without</i>	<i>With</i>	<i>Without</i>	<i>With</i>	<i>Without</i>	<i>With</i>
	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>	<i>Controls</i>
Posterior belief	-0.002*** (0.001)	-0.001** (0.000)	-0.013*** (0.003)	-0.007** (0.003)	-0.671*** (0.165)	-0.359** (0.149)
Control mean	0.160	0.160	0.712	0.712	34.148	34.148
Observations	714	714	714	714	714	714
Controls	No	Yes	No	Yes	No	Yes

Notes: Table A13 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing the indicated behavioral outcomes on control group respondents' beliefs about the share of competitor party members who will go canvassing during the campaign (posterior belief). Pre-specified control variables include: party membership, number of years of party membership, age, gender, whether a participant has participated in a canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this election and whether a participant has canvassed in the current election. The sample is restricted to the control group. *p<0.1; **p<0.05; ***p<0.01.

Table A14: Effects on social media activity

	Shared social media message		
	Any	Days	Total
Treatment	-0.004 (0.012)	0.130 (0.115)	0.297 (0.222)
Control mean	0.075	0.306	0.703
Observations	1417	1417	1417

Notes: Table A14 presents treatment effects on social media activity. “*Any*” takes value one if the respondent shares any party news story on Facebook through the application. “*Days*” denotes the total number of days a respondent shares a party news story on Facebook through the application. “*Total*” is the total number of party news stories shared by the respondent on Facebook through the application. Treatment effects are obtained conditional on pre-specified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

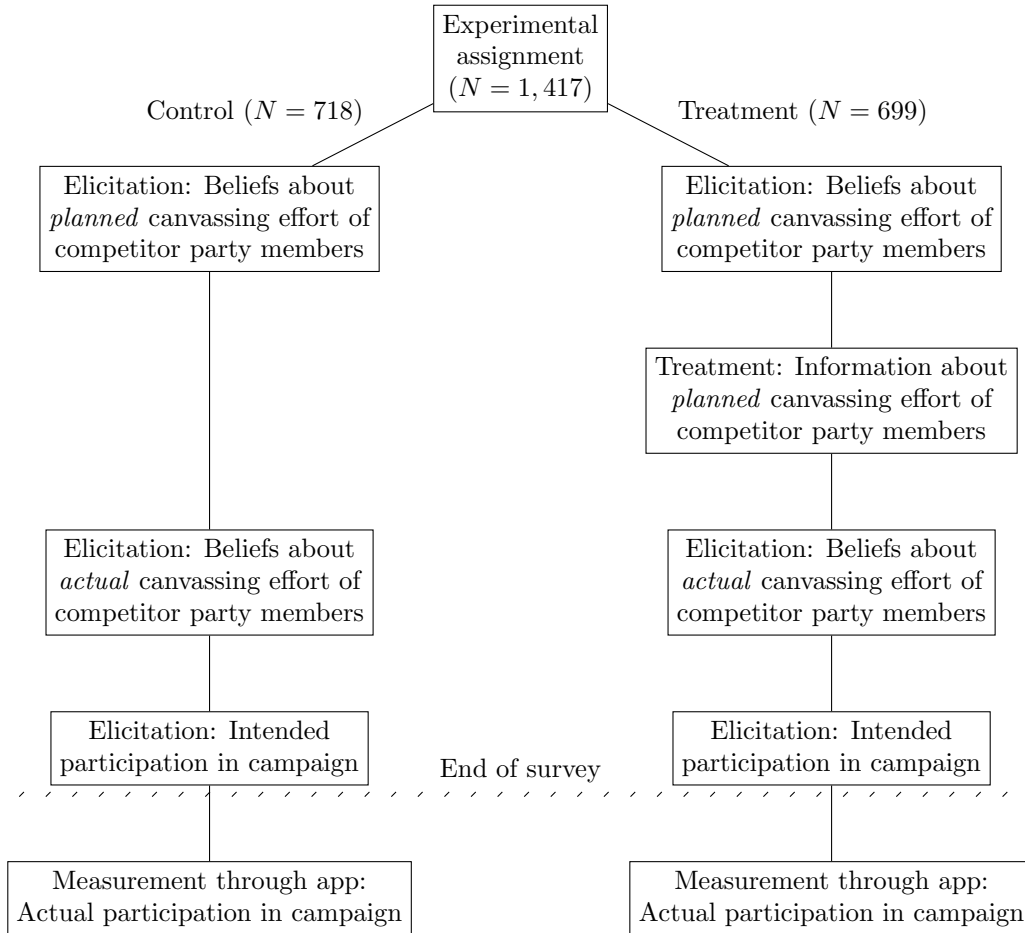
Table A15: Effects on social media activity (underestimators only)

	Shared social media message		
	Any	Days	Total
Treatment	-0.004 (0.013)	0.132 (0.123)	0.295 (0.242)
Control mean	0.078	0.329	0.750
Observations	1271	1271	1271

Notes: Table A15 presents treatment effects on social media activity for respondents who underestimate the share of competitor party members who plan to participate in the party's door-to-door canvassing campaign. "*Any*" takes value one if the respondent shares any party news story on Facebook through the application. "*Days*" denotes the total number of days a respondent shares a party news story on Facebook through the application. "*Total*" is the total number of party news stories shared by the respondent on Facebook through the application. Treatment effects are obtained conditional on pre-specified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Additional Figures

Figure A.1: Experimental design

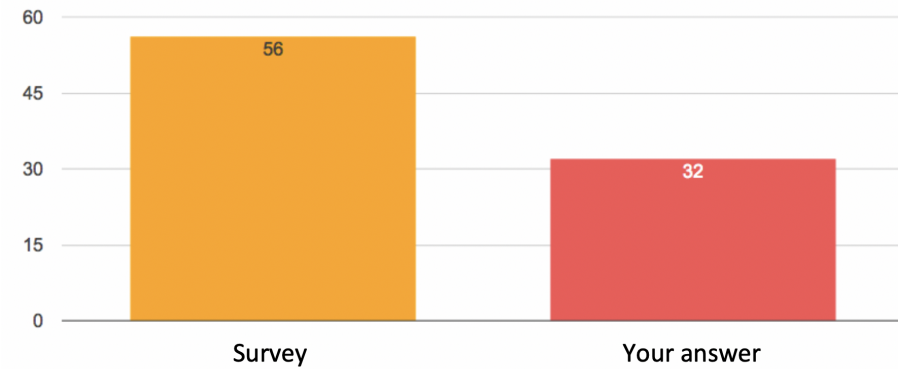


Notes: Figure A.1 illustrates the experimental design.

Figure A.2: Treatment screen

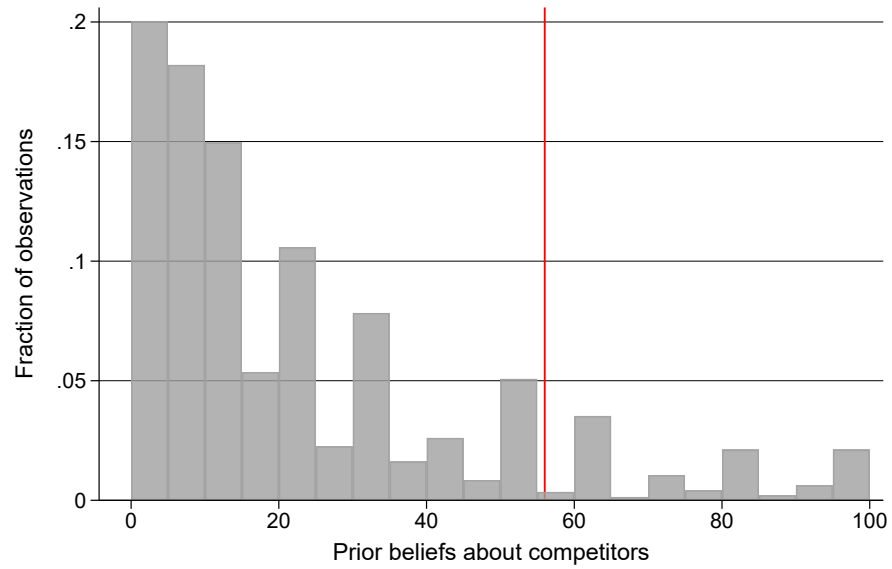
You said **32 of 100 [competitor] party members**.

According to a survey of [competitor] party members, **56 of 100 [competitor] party members** plan to go canvassing during this electoral campaign.



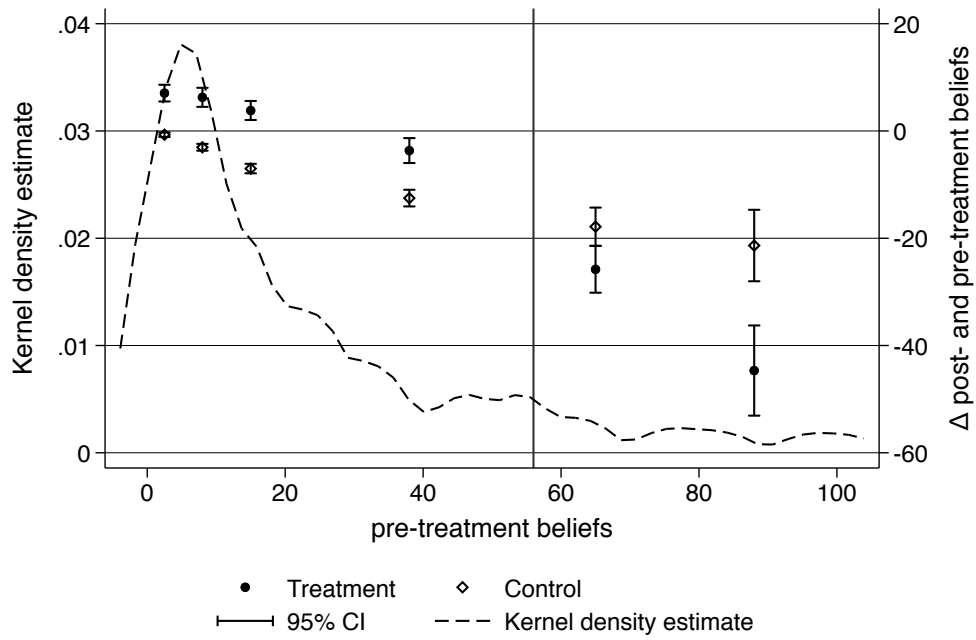
Notes: Figure A.2 displays an exemplary treatment screen for a respondent who believes that 32 percent of members of the competitor party will go canvassing.

Figure A.3: Distribution of pre-treatment beliefs about competitor effort



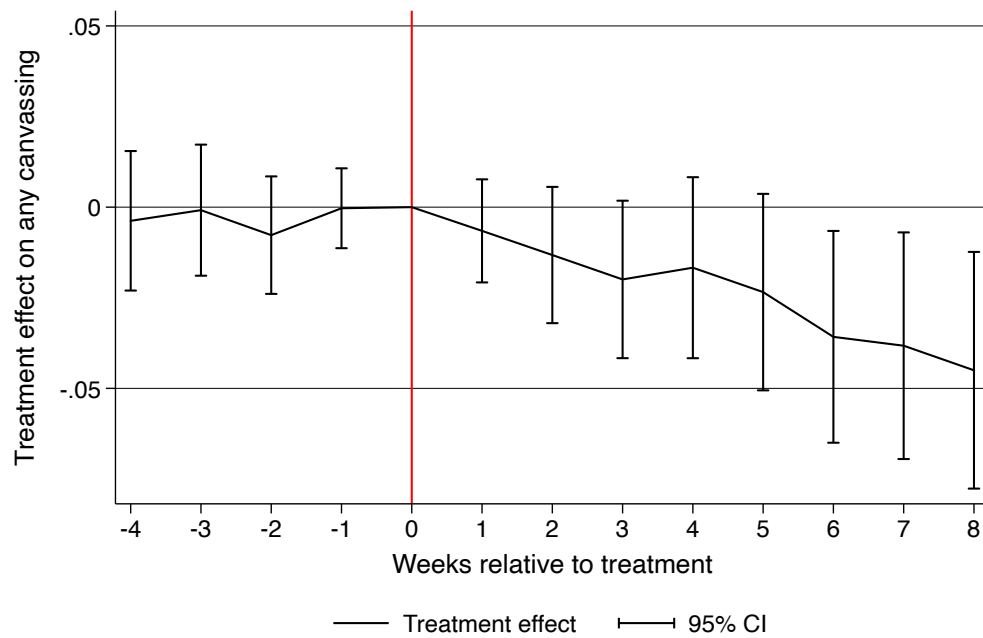
Notes: Figure A.3 shows a histogram of pre-treatment beliefs about the fraction of competitor party members who plan to go canvassing. The vertical red line (56 percent) corresponds to the treatment information.

Figure A.4: Belief changes over pre-treatment beliefs



Notes: Figure A.4 shows average difference between post- beliefs and pre-treatment beliefs the fraction of members of the competitor party who plan to go canvassing in the treatment and control group. The vertical line (56 percent) corresponds to the true treatment information, which is based on survey evidence. Respondents are grouped according to treatment beliefs in six bins. Respondents who underestimate the treatment information are grouped in quartiles. Respondents who underestimate the treatment information are grouped in below and above median pre-treatment beliefs.

Figure A.5: Treatment effects on any canvassing over time



Notes: Figure A.5 plots the treatment effects on any canvassing over time using a difference-in-differences specification that controls for individual fixed effects. The sample is restricted to underestimators. The black vertical lines indicate 95% confidence intervals. The vertical red line indicates the timing of the treatment. The election took place in week eight.

Heterogeneity

Did the treatment yield greater behavioral changes among specific subgroups? To streamline the heterogeneity analysis, we focus on treatment-by-covariate interactions for respondents' participation decision. Using the covariate data at our disposal, this allows us to scrutinize eight subgroups. Four of these subgroups were pre-registered: respondents that had downloaded the app (Table A16), respondents with prior canvassing experience (Table A17), respondents that are party members (Table A18), and respondents' years of membership (Table A19). For the sake of completeness, we report the effect of all potential interactions on the participation decision in Table A20.

Table A16: Treatment effects (app heterogeneity)

	Posterior	Intentions		App Data		
	Belief	Any	Days	Any	Days	Door
Panel A: No app download						
Treatment	5.119*** (1.146)	0.043 (0.027)	0.158 (0.338)	-0.017 (0.011)	-0.031 (0.068)	-3.076 (3.696)
Control mean	16.454	0.384	2.722	0.044	0.143	7.484
Observations	1,009	1,006	1,006	1,017	1,017	1,017
Panel B: App download						
Treatment	5.513*** (1.667)	0.037 (0.034)	0.975 (0.833)	-0.065 (0.047)	0.007 (0.444)	12.961 (23.072)
Control mean	12.750	0.839	7.087	0.435	2.051	96.944
Observations	397	396	396	400	400	400

Notes: Table A16 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing all available outcome variables on the treatment dummy for the indicated sample. Treatment effects are obtained conditional on pre-specified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in canvassing training, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Table A17: Treatment effects (experience heterogeneity)

	Posterior	Intentions		App Data		
	Belief	Any	Days	Any	Days	Door
Panel A: Inexperienced supporters						
Treatment	6.114*** (1.240)	0.012 (0.027)	0.560* (0.335)	-0.030 (0.018)	-0.012 (0.142)	1.944 (7.577)
Control mean	15.090	0.407	2.736	0.130	0.487	23.785
Observations	881	877	877	888	888	888
Panel B: Experienced supporters						
Treatment	3.767** (1.468)	0.096*** (0.036)	-0.071 (0.675)	-0.036 (0.028)	-0.048 (0.271)	-1.735 (13.921)
Control mean	15.786	0.701	6.097	0.210	1.081	51.140
Observations	525	525	525	529	529	529

Notes: Table A17 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing all available outcome variables on the treatment dummy for the indicated sample. Treatment effects are obtained conditional on pre-specified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election.

*p<0.1; **p<0.05; ***p<0.01.

Table A18: Treatment effects (membership heterogeneity)

	Posterior	Intentions		App Data		
	Belief	Any	Days	Any	Days	Door
Panel A: No member						
Treatment	1.777 (2.551)	-0.067 (0.052)	-0.478 (0.598)	0.009 (0.031)	0.051 (0.158)	-0.810 (8.978)
Control mean	18.609	0.381	2.929	0.078	0.267	15.948
Observations	246	244	244	251	251	251
Panel B: Party member						
Treatment	5.757*** (1.013)	0.071*** (0.024)	0.584 (0.391)	-0.041** (0.018)	-0.065 (0.160)	0.220 (8.360)
Control mean	14.730	0.545	4.223	0.176	0.797	37.654
Observations	1,160	1,158	1,158	1,166	1,166	1,166

Notes: Table A18 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing all available outcome variables on the treatment dummy for the indicated sample. Treatment effects are obtained conditional on pre-specified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Table A19: Treatment effects (membership years heterogeneity)

	Posterior	Intentions		App Data		
	Belief	Any	Days	Any	Days	Door
Panel A: Below						
med. membership dur.						
Treatment	6.714*** (1.288)	-0.001 (0.030)	0.268 (0.449)	-0.040 (0.024)	0.088 (0.222)	5.724 (11.577)
Control mean	14.561	0.570	4.503	0.206	0.873	43.285
Observations	737	735	735	744	744	744
Panel B: Above						
med. membership dur.						
Treatment	3.391** (1.394)	0.090*** (0.032)	0.430 (0.487)	-0.027 (0.019)	-0.161 (0.158)	-5.434 (8.063)
Control mean	16.237	0.463	3.478	0.109	0.531	23.932
Observations	669	667	667	673	673	673

Notes: Table A19 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing all available outcome variables on the treatment dummy for the indicated sample. Treatment effects are obtained conditional on pre-specified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Table A20: Treatment effect heterogeneity

	Any canvassing							
Treatment	-0.027 (0.018)	-0.067* (0.039)	0.009 (0.031)	-0.048** (0.021)	-0.031* (0.018)	-0.019 (0.015)	-0.032** (0.015)	-0.017 (0.011)
T × Female	-0.021 (0.037)							
T × Age		0.001 (0.001)						
T × Member			-0.050 (0.036)					
T × Membership years				0.001 (0.001)				
T × Experience					-0.004 (0.033)			
T × Canvassing workshop						-0.062 (0.048)		
T × Has canvassed this election							-0.005 (0.084)	
T × Downloaded app before survey								-0.055 (0.048)
R-squared	0.284	0.285	0.285	0.285	0.284	0.286	0.284	0.285
Observations	1,417	1,417	1,417	1,417	1,417	1,417	1,417	1,417

Notes: Table A20 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing the indicated behavioral outcome (any canvassing) on the treatment dummy interacted with the indicated covariates for the full sample. Treatment effects are obtained conditional on pre-specified control variables: party membership, number of years of party membership, age, gender, whether a participant has participated in canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Theoretical model

Set-up

How does party competition affect a party supporter's decision to become politically active? To structure our thinking, consider a potential activist i whose utility depends on the outcome of an election with two competing parties. The utility gained from the election is a function of the number of votes $v(\cdot)$, which is described by $g(v(d_i + d_{-i}, cd))$. $g(\cdot)$ determines how effective votes are in winning the election or reaching the political outcome. The vote share for i 's party, $v(d_i + d_{-i}, cd)$, depends on her canvassing effort level, $d_i \geq 0$, the effort of other activists in i 's party (d_{-i}), as well as the canvassing level of the competitor party cd . We further allow for expressive utility $k(d_i, cd)$ which captures non-instrumental benefits. Finally, we include individual-specific fixed costs of engaging in any canvassing $\tilde{c}_i$, which are distributed according to F , and continuous costs as a function of canvassing effort $c(d_i)$. Following this simple setup, activist i 's utility can be expressed as follows:

$$u_i(d_i, d_{-i}, cd) = g(v(d_i + d_{-i}, cd)) + k(d_i, cd) - c(d_i) - \tilde{c}_i 1(d_i > 0) \quad (1)$$

Canvassing effort choice

Conditional on participating in canvassing, i chooses the canvassing effort according to the following FOC:

$$\frac{du_i(d_i, d_{-i}, cd)}{dd_i} = \frac{\partial g(v(d_i^* + d_{-i}, cd))}{\partial v} \frac{\partial v(d_i^* + d_{-i}, cd)}{\partial d_i} + \frac{\partial k(d_i^*, cd)}{\partial d_i} - \frac{dc(d_i^*)}{dd_i} = 0 \quad (2)$$

From equation 2 we see that i 's effort choice depends on the marginal utility gained from vote shares, the effectiveness of her canvassing actions in gaining additional vote shares, and on her expressive benefits. To classify the response to an increase in competition, we consider the following high-level assumptions.

1. Rational: people care about winning the election.
2. Effectiveness: canvassing gains you votes.
3. Concavity: the marginal utility of additional vote shares is decreasing. This corresponds to the assumption that the party is ahead in the polls (which

was the case in our setting). Furthermore, the marginal impact on voter persuasion decreases in each additional unit of canvassing effort.

4. Change in effectiveness: canvassing by the opposing party changes the effectiveness of one's own canvassing.
5. Expressive motives: competition can either increase or decrease the expressive value of own canvassing.
6. Self-limiting expressive motives: own canvassing does not increase the expressive value of further own canvassing. If that were not the case, canvassers might motivate themselves to knock on infinitely many doors.

With these assumptions and implicit differentiation we get an ambiguous theoretical prediction about the impact of an increase in competition on the canvassing effort decision.

$$\begin{aligned}
\frac{dd_i^*}{dcd} = & - \frac{
\begin{array}{c}
\text{>0, this is the closeness effect} \\
\overbrace{\frac{\partial^2 g(v(\cdot))}{(\partial v(\cdot))^2} \frac{\partial v(\cdot)}{\partial cd} \frac{\partial v(\cdot)}{\partial d_i}}^{<0, \text{ by 3. } <0, \text{ by 2. } >0, \text{ by 2.}} + \\
\frac{\partial^2 g(v(\cdot))}{(\partial v(\cdot))^2} \frac{\partial v(\cdot)}{\partial cd} \frac{\partial v(\cdot)}{\partial d_i} + \\
\text{<0, by 3. } >0, \text{ quadratic}
\end{array}
+ \frac{
\begin{array}{c}
\text{<0 if less effective, this is the decreased effectiveness} \\
\overbrace{\frac{\partial g(v(\cdot))}{\partial v(\cdot)} \frac{\partial^2 v(\cdot)}{\partial d_i \partial cd}}^{>0, \text{ by 1. } <0, \text{ by 4.}} \\
\frac{\partial g(v(\cdot))}{\partial v(\cdot)} \frac{\partial^2 v(\cdot)}{\partial d_i \partial cd} + \\
\text{>0, by 1. } \leq 0, \text{ by 3. }
\end{array}
+ \frac{
\begin{array}{c}
0 \leq, \text{ by 5.} \\
\frac{\partial^2 k(d_i, cd)}{\partial d_i \partial cd} \\
0 \leq, \text{ by 6. }
\end{array}
}{
\begin{array}{c}
\frac{\partial^2 g(\cdot)}{(\partial v(\cdot))^2} \left(\frac{\partial v(\cdot)}{\partial d_i} \right)^2 + \frac{\partial g(v(\cdot))}{\partial v(\cdot)} \frac{\partial v(\cdot)}{\partial d_i \partial d_i} + \frac{\partial^2 k(d_i, cd)}{(\partial d_i)^2} - \frac{d^2 c(d_i^*)}{(dd_i)^2} \\
\text{<0, by 3. } >0, \text{ quadratic} \quad \text{>0, by 1. } \leq 0, \text{ by 3.} \quad 0 \leq, \text{ by 6.} \quad \geq 0 \\
\text{<0} \quad \text{\leq 0} \quad \text{0 \leq}
\end{array}
}
\end{aligned}$$

The denominator is unambiguously negative, which means that the sign of the numerator determines the nature of the strategic interactions. If instrumental motives do not play a role, it is only the nature of expressive motives that determines the whether competition increases or decreases canvassing activity. If instrumental concerns do play a role, the closeness and reduced effectiveness effects also play a role in determining the sign of the interaction. If the closeness effect is strongest an increase in competition will lead to an increase in canvassing activity. However, if effectiveness is decreasing in higher competitor effort, the effect of decreased effectiveness will lead to a decrease in canvassing activity. Finally, it is also conceivable that effectiveness increases in competitor effort, which will reinforce a positive effect of competition on canvassing activity.

Participation decision

To investigate the impact of changes in competition on the participation decision, we explore the role of fixed cost $\tilde{c}_i$. The fraction of individuals going canvassing at a given optimal canvassing level d^* is determined by the difference in utility between participating and non-participating. The marginal activist should just be indifferent between the two.

$$\begin{aligned}\Delta u &= u(d^*, d_{-i}, cd) - u(0, d_{-i}, cd) = g(v(d_i^* + d_{-i}, cd)) + k(d_i^*, cd) - c(d_i^*) - \tilde{c}_i \\ &\quad - g(v(d_{-i}, cd)) - k(0, cd) = 0\end{aligned}$$

We assume that $k(0, cd) = 0 \forall cd$, that is individuals do not gain any expressive utility if they do not canvass. Using the distributional assumption on $\tilde{c}_i$, we obtain the following equation for the fraction of active canvassers.

$$F(g(v(d_i^* + d_{-i}, cd)) + k(d_i^*, cd) - c(d_i^*) - g(v(d_{-i}, cd))) \quad (3)$$

To explore the impact on the participation decision, we take the first derivative of F with respect to cd .

$$\begin{aligned}\frac{dF}{dcd} &= \left[\frac{\partial g(v(d_i^* + d_{-i}, cd))}{\partial v} \left(\frac{\partial v(d_i^* + d_{-i}, cd)}{\partial cd} + \frac{\partial v(d_i^* + d_{-i}, cd)}{\partial d_i^*} \frac{\partial d_i^*}{\partial cd} \right) \right. \\ &\quad \left. + \frac{\partial k(d_i^*, cd)}{\partial cd} + \frac{\partial k(d_i^*, cd)}{\partial d_i^*} \frac{\partial d_i^*}{\partial cd} - \frac{dc(d_i^*)}{dd_i^*} \frac{\partial d_i^*}{\partial cd} - \frac{g(v(d_{-i}, cd))}{\partial v} \frac{\partial v(d_{-i}, cd)}{\partial cd} \right] \\ &\quad \cdot f(\cdot)\end{aligned}$$

Using the first order condition 2, this can be simplified to:

$$\begin{aligned}\frac{dF}{dcd} &= \left[\frac{\partial g(v(d_i^* + d_{-i}, cd))}{\partial v} \frac{\partial v(d_i^* + d_{-i}, cd)}{\partial cd} + \frac{\partial k(d_i^*, cd)}{\partial cd} - \frac{g(v(d_{-i}, cd))}{\partial v} \frac{\partial v(d_{-i}, cd)}{\partial cd} \right] \\ &\quad \cdot f(\cdot)\end{aligned}$$

Given our previous assumptions on concavity, we know that $0 < \frac{\partial g(v(d_i^* + d_{-i}, cd))}{\partial v} < \frac{\partial g(v(d_{-i}, cd))}{\partial v}$. However, the relative effect of competition on the party's own vote share is ambiguous $\frac{\partial v(d_i^* + d_{-i}, cd)}{\partial cd} \gtrless \frac{\partial v(d_{-i}, cd)}{\partial cd} < 0$. Hence, the sign of the impact on the

participation decision depends on the sum of two factors. First, the sign depends on whether the effect of competition on additional votes generate (effectiveness) increases or decreases in own canvassing. Second, the sign and magnitude of the direct impact of competition on the expressive utility. Finally, the magnitude of the effect is pinned down by the density function at the fixed cost value of the marginal (or indifferent) canvasser.

Demand effects

In this section, we discuss one methodological facet of our study. As stated, in addition to collecting rich behavioral data, the original survey also included two self-reported outcomes. Specifically, we asked respondents whether they planned to go canvassing and, if so, on how many days. Do the behavioral findings map onto respondents' self-reported intentions?²⁴

In Table A21, we report treatment effects on respondents' intentions to go canvassing. In contrast to the behavioral finding, we do *not* find a negative treatment effect on canvassing intentions. If anything, there is a slight positive effect (though the coefficient is insignificant without control variables). This short-term effect may be a product of social pressure induced by the surveyor: Respondents might want to please the party in reaction to learning about high levels of competitor effort. In reality, however, they dropped out of the campaign. These results underscore the necessity to collect unobtrusive behavioral data in order to reliably measure political engagement. Relying on self-reported intentions to predict engagement would have led to the wrong conclusion, namely, that the competition treatment mildly increased engagement.

²⁴In the pre-analysis plan, we pre-specified an index, which combines the behavioral and survey data. We had anticipated that intentions and behaviors point into the same direction. But, given that the survey responses are at odds with behavior, the index is of little empirical use.

Table A21: Treatment effects on self-reported intentions

	Planned canvassing		Planned days	
	(1)	(2)	(3)	(4)
Treatment	0.027 (0.027)	0.046** (0.022)	0.235 (0.378)	0.418 (0.340)
Control mean	0.519	0.519	4.017	4.017
Observations	1,402	1,402	1,402	1,402
Controls	No	Yes	No	Yes

Notes: Table A21 presents coefficients and robust standard errors (in parentheses) of a linear model (OLS), regressing the indicated self-reported intentions on the treatment dummy for the full sample. “Planned canvassing” is a dummy indicating if respondents plan to go canvassing. “Planned days” is a continuous variable measuring the number of days respondents plan to go canvassing. Pre-specified control variables include: party membership, number of years of party membership, age, gender, whether a participant has participated in a canvassing training, whether a participant has already downloaded the online application, whether a participant has participated in canvassing before this national election and whether a participant has canvassed in the current election. *p<0.1; **p<0.05; ***p<0.01.

Survey instrument

- **Introduction**

Dear [name],

We are conducting a short survey among our supporters to plan our electoral campaign. Your participation helps us optimally use our campaign resources. We will treat your answers confidentially. The survey only takes 5 minutes (10 questions).

Thank you very much for your help!

- **Gender**

What is your gender?

- **Age**

How old are you?

- **Party member**

Are you a member of [own] party?

- **Years of party membership** (asked if respondent is a member)

For how many years have you been a member of [own] party?

- **Canvassing workshop**

Have you ever participated in a canvassing training workshop?

- **Canvassing experience**

Do you have any experience in canvassing in previous electoral campaigns?

- **Prior belief**

Think of 100 typical [competitor name] party members.

What do you think: How many of these 100 [competitor name] party members plan to go canvassing during this electoral campaign?

- **Treatment text** (randomly assigned)

You said X of 100 [competitor name] party members.

According to a survey of [competitor name] party members, 56 of 100 [competitor name] party members plan to go canvassing during this electoral campaign.

- **Posterior belief**

What do you think: How many of these 100 [competitor] party members will **actually** go canvassing during this electoral campaign?

- **Participation decision**

Do you plan to go canvassing during this electoral campaign?

- **Effort decision** (asked if participation decision is ‘yes’)

On how many days do you plan to go canvassing during this electoral campaign?

- **Debrief**

Now let’s go! And don’t forget to download the [party]-application. Available here for iOS and Android.

With the [party]-application you can actively participate in our electoral campaign and keep up to date with the campaign progress. Also, the application is fun!